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than the sapwood. The timber from the heartwood is more durable and more resistant to the attack of microorganisms and insects than the timber from sapwood.



Figure 10.10: Cross section of natural wood

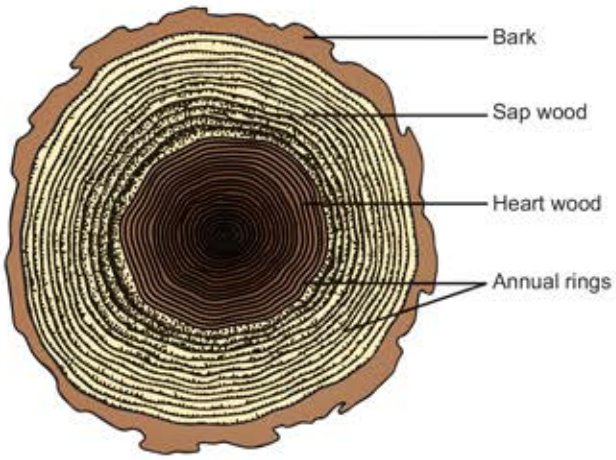


Figure 10.11: Cross - section of wood showing annual ring



When, the heart wood of a tree is destroyed, no vital function of the plant is affected.

When, the sap wood is destroyed, the plant will die because conduction of water will be blocked.

Differences Between Sap Wood (alburnum) and Heart Wood (duramen)

Sap Wood (Alburnum)	Heart Wood (Duramen)
Living part of the wood.	Dead part of the wood.
It is situated on the outer side of wood	It is situated in the centre part of wood
It is less in coloured	It is dark in coloured
Very soft in nature	Hard in nature
Tyloses are absent	Tyloses are present
It is not durable and not resistant to microorganisms	It is more durable and resists microorganisms



The dye, haematoxylin is obtained from the heart wood of *Haematoxylum campechianum* used to stain plant materials for observation under microscope, especially the nucleus of the cell.

a. Haematoxylin



Abies balsamea is a gymnospermic plant. It produces canada balsam, from its resin ducts. It is used as mounting medium for microscopic slide preparation.

b. Canada balsam